

Ethan Karpeles

Dallas, Texas, USA

www.linkedin.com/in/ethankarpeles

PROFESSIONAL PROFILE

AI & Software Engineer at PMG and award-winning Mathematics graduate from the University of North Texas. Bilingual in English and Spanish.

PROGRAMMING LANGUAGES

Python | SQL | C++ | C | JavaScript | Java | HTML | CSS | LaTeX

SOFTWARE

Git | GNU/Linux | Excel

LANGUAGES

English (Native) | Spanish (Bilingual)

EXPERIENCE

PMG Dallas, Texas, USA
2026 – Present

AI & Software Engineer I

- Develops and maintains a full-stack data categorization tool that empowers analysts to efficiently process and organize complex datasets.
- Participated in the Graduate Leadership Program for AI & Software Engineering.

AI & Software Engineering Intern

Summer 2025

- Deployed AI-driven dashboards, flagging major service issues and performance bottlenecks for proactive engineering resolution. Presented the final product to senior executives.
- Developed automated Jira issue ingestion workflows and integrated OpenAI embeddings with KMeans clustering to enable ML-based ticket classification.
- Engineered and documented Python scripts for BigQuery table management, streamlining internal data pipelines and eliminating manual credential hurdles.

EDUCATION

Bachelor of Science in Mathematics

Minor in Computer Science

University of North Texas, Denton, Texas, USA

GPA: 4.0
2021 – 2025

RELEVANT COURSEWORK

Mathematics

Numerical Analysis | Graph Theory | Linear Algebra | Applied Statistics | Differential Equations | Calculus I–III | Graduate-Level Elliptic Curve Seminar | Real Analysis I and II | Abstract Algebra | Topology

Computer Science

Cryptography (Python & C++) | Applied Artificial Intelligence (Python) | Foundations of Data Structures (C++) | Foundations of Computing | Computer Science I and II (C++)

HONORS & AWARDS

Outstanding Undergraduate Mathematics Student of the Year, UNT Math Department

2024

President's List

Spring 2024

RESEARCH EXPERIENCE

University of North Texas

Denton, Texas, USA

Undergraduate Research Fellowship

2024 – 2025

- Wrote a Python program to identify families of elliptic curves with semistable reduction modulo any prime.
- Gave a talk in a full auditorium for the entire math department and presented a poster at UNT's Scholars Day.

LEADERSHIP EXPERIENCE

President, Alpha Epsilon Pi, UNT

2024 – 2025

- Elected to lead a 20-member chapter, overseeing operations, recruitment, community service initiatives, and the management of a \$28,000 annual budget.

Vice President, Math Club, UNT

2024 – 2025

- Assisted in the leadership of an academic organization to foster a deeper interest in mathematics for students.